Лабораторная 4

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CREATE TABLE car_class (
   class_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
   class_name TEXT NOT NULL UNIQUE,
   class_fare FLOAT NOT NULL CHECK (class_fare > 0)
);

CREATE TABLE facility (
   facility_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
   facility_name TEXT NOT NULL UNIQUE,
   facility_cost FLOAT NOT NULL CHECK (facility_cost > 0)
);

CREATE TABLE payment_type (
   type_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
   type_name TEXT NOT NULL UNIQUE
);

CREATE TABLE user_role (
   role_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
   role_name TEXT NOT NULL UNIQUE
);

CREATE TYPE gender_type AS ENUM ('male', 'female');

CREATE TABLE "user" (
   user_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
   user_login TEXT NOT NULL UNIQUE,
   password_hash CHAR(64) NOT NULL,
   passport_number CHAR(6) NOT NULL CHECK (passport_number ~ '^[0-9]{6}$'),
   passport_series CHAR(4) NOT NULL CHECK (passport_series ~ '^[0-9]{4}$'),
   full_name TEXT NOT NULL,
   phone_number CHAR(11) NOT NULL CHECK (phone_number ~ '^8[0-9]{10}$'),
   birth_date DATE NOT NULL CHECK (
      birth_date <= CURRENT_DATE - INTERVAL '18 years'
      AND birth_date >= CURRENT_DATE - INTERVAL '100 years'
   ),
   role_id INT NOT NULL REFERENCES user_role (role_id) ON DELETE RESTRICT,
   gender gender_type NOT NULL,
   CONSTRAINT uq_passport UNIQUE (passport_number, passport_series)
);

CREATE TABLE car_brand (
   brand_id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
   brand_name TEXT NOT NULL UNIQUE
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);

CREATE TABLE car_model (
    model_id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
    model_name TEXT NOT NULL UNIQUE,
    brand_id BIGINT NOT NULL REFERENCES car_brand (brand_id) ON DELETE RESTRICT
);

CREATE TABLE car (
    car_id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
    car_number CHAR(9) NOT NULL,
    release_year INT NOT NULL CHECK (release_year > 1900 AND release_year < 2100),
    rent_cost FLOAT NOT NULL CHECK (rent_cost > 0),
    maintenance_cost FLOAT NOT NULL CHECK (maintenance_cost > 0),
    property BOOLEAN NOT NULL,
    brand_id INT NOT NULL REFERENCES car_brand (brand_id) ON DELETE RESTRICT,
    model_id INT NOT NULL REFERENCES car_model (model_id) ON DELETE RESTRICT
);

CREATE TABLE contract (
    cotract_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    "start_date" DATE NOT NULL,
    end_date DATE NOT NULL CHECK ("start_date" < end_date),
    manager_id BIGINT NOT NULL REFERENCES "user" (user_id) ON DELETE RESTRICT,
    driver_id BIGINT NOT NULL REFERENCES "user" (user_id) ON DELETE RESTRICT,
    car_id BIGINT NOT NULL REFERENCES car (car_id) ON DELETE RESTRICT
    -- CONSTRAINT chk_manager_role CHECK (
    --     EXISTS (
    --         SELECT 1 FROM user u
    --         JOIN user_role r ON u.role_id = r.role_id
    --         WHERE u.user_id = manager_id
    --         AND r.role_name = 'manager'
    --     )
    -- ),
    -- CONSTRAINT chk_driver_role CHECK (
    --     EXISTS (
    --         SELECT 1 FROM user u
    --         JOIN user_role r ON u.role_id = r.role_id
    --         WHERE u.user_id = driver_id
    --         AND r.role_name = 'driver'
    --     )
    -- )
);

CREATE TABLE car_facility (
    facility_id INT NOT NULL REFERENCES facility (facility_id) ON DELETE RESTRICT,
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    car_id BIGINT NOT NULL REFERENCES car (car_id) ON DELETE RESTRICT,
    CONSTRAINT pk_facility_car PRIMARY KEY (facility_id, car_id)
);

CREATE TABLE "order" (
    order_id BIGINT NOT NULL,
    driver_id BIGINT NOT NULL REFERENCES "user" (user_id) ON DELETE RESTRICT,
    location_from TEXT NOT NULL,
    location_to TEXT NOT NULL,
    order_time TIMESTAMP NOT NULL,
    completed BOOLEAN NOT NULL,
    start_time TIMESTAMP,
    end_time TIMESTAMP CHECK (
        NOT completed AND start_time IS NULL AND end_time IS NULL
        OR
        completed AND start_time IS NOT NULL AND start_time < end_time AND order_time <
start_time
    ),
    passenger_id BIGINT NOT NULL REFERENCES "user" (user_id) ON DELETE RESTRICT,
    car_id BIGINT NOT NULL REFERENCES car (car_id) ON DELETE RESTRICT,
    CONSTRAINT pk_order_driver PRIMARY KEY (order_id, driver_id)
    -- CONSTRAINT chk_driver_role CHECK (
    --     EXISTS (
    --         SELECT 1 from "user" u
    --         JOIN user_role r ON u.role_id = r.role_id
    --         WHERE u.user_id = driver_id
    --         AND r.role_name = 'driver'
    --     )
    -- ),
    -- CONSTRAINT chk_passenger_role CHECK (
    --     EXISTS (
    --         SELECT 1 from "user" u
    --         JOIN user_role r ON u.role_id = r.role_id
    --         WHERE u.user_id = passenger_id
    --         AND r.role_name = 'passenger'
    --     )
    -- ),
    -- CONSTRAINT chk_driver_car_possesion CHECK (
    --     EXISTS (
    --         SELECT 1 FROM contract c
    --         JOIN user u ON c.driver_id = u.user_id
    --         JOIN car ON c.car_id = car.car_id
    --         WHERE car_id = c.car_id
    --         AND driver_id = c.driver_id
    --         AND order_time > c.start_date
    --         AND order_time < c.end_date
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    --     )
    -- )
);

CREATE TABLE review (
    order_id BIGINT NOT NULL,
    driver_id BIGINT NOT NULL,
    user_id BIGINT NOT NULL,
    "content" TEXT,
    rate INT NOT NULL CHECK (rate >= 1 AND rate <= 5),
    FOREIGN KEY (order_id, driver_id) REFERENCES "order" (order_id, driver_id) ON DELETE
RESTRICT,
    CONSTRAINT pk_order_user PRIMARY KEY (order_id, user_id)
    -- CONSTRAINT chk_order_is_completed (
    --     EXISTS (
    --         SELECT 1 FROM "order" o
    --         WHERE order_id = o.order_id
    --         AND o.completed
    --     )
    -- ),
    -- CONSTRAINT chk_user_is_passenger_or_driver (
    --     EXISTS (
    --         SELECT 2 FROM "order" o
    --         WHERE order_id = o.order_id
    --         AND (
    --             user_id = o.driver_id
    --             OR user_id = o.passenger_id
    --         )
    --     )
    -- ),
);

CREATE TABLE payment (
    payment_id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,
    type_id INT NOT NULL REFERENCES payment_type (type_id) ON DELETE RESTRICT,
    payment_time TIMESTAMP NOT NULL,
    car_id BIGINT REFERENCES car (car_id) ON DELETE RESTRICT,
    order_id BIGINT,
    driver_id BIGINT,
    FOREIGN KEY (order_id, driver_id) REFERENCES "order" (order_id, driver_id) ON DELETE
RESTRICT,
    -- выплата должна рассчитываться на основе других значений
    amount FLOAT NOT NULL,
    CONSTRAINT chk_car_order_elimination CHECK (
        car_id IS NULL AND order_id IS NOT NULL
        OR car_id IS NOT NULL AND order_id IS NULL
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    )
);

CREATE TABLE ordered_facility (
    facility_id INT NOT NULL REFERENCES facility (facility_id) ON DELETE RESTRICT,
    order_id BIGINT NOT NULL,
    driver_id BIGINT NOT NULL,
    FOREIGN KEY (order_id, driver_id) REFERENCES "order" (order_id, driver_id) ON DELETE
RESTRICT
);
```